

UNIVERSITY OF CAPE COAST

**EFFECT OF GEOGEBRA SOFTWARE ON STUDENTS'
ACHIEVEMENT IN DETERMINING MINIMUM AND MAXIMUM
VALUES OF QUADRATIC FUNCTION AT THE SENIOR HIGH
SCHOOL LEVEL**

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2024

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BY

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This project was submitted to the Department of Mathematics and ICT
Education at the College of Education Studies, University of Cape Coast, as
partially fulfilment of the requirements for the Bachelor of Education degree
in Mathematics Education

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SEPTEMBER 2024

DECLARATION

Candidates' Declaration

We hereby declare that this project work is the result of our own original research and that no part of it has been presented for degree in this university or elsewhere.

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Supervisor's Declaration

I hereby confirm that the project work was prepared and presented under supervision in compliance with the University of Cape Coast's defined supervision criteria.

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ABSTRACT

The study investigated the impact of the GeoGebra software on students' ability to determining the minimum and maximum values of quadratic functions at the senior high school level. The study utilized purposive sampling to select two senior high schools, with a sample size of 75 students from each school. It employed a quasi-experimental design., with an identical set of students receiving quadratic function instruction via the conventional way and the GeoGebra program. Pre-test and post-test instruments were used to compare the performance of the identical set of students before and after they were taught using GeoGebra. The pre-test and post-test scores were analysed using mean, standard deviation and t-test. The results revealed that GeoGebra intervention was effective in raising the overall performance of the participants. Additionally, a questionnaire comprising 29 items was administered to students after the intervention to collect data on students' attitude, motivation, and acceptance of the GeoGebra instruction. Percentages were used to analyse the questionnaire responses. The findings showed GeoGebra helped enhance the students understanding of these concepts and gained confidence in identifying the minimum and maximum values of quadratic functions after being taught using GeoGebra. Therefore, most students viewed the software as a valuable tool for learning mathematics.

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DEDICATION

To our loved ones.

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CHAPTER ONE INTRODUCTION

Mathematics is the language that enables us to describe ideas and relationships derived from our surroundings (Kitta, 2004). According to Salamah (2003), “mathematics is not just a static collection of knowledge, but a dynamic activity that involves problem solving, reasoning, and communication”. Mathematics education is one of the essential pillars within the educational landscape as it is an essential factor in the human life whether been in the case of student life, professional or business life.

Quadratic functions, a Sub-strand of algebra is one of the topics crucial for understanding patterns and relationships in mathematical models and realworld scenarios. The study of the characteristics of quadratic functions helps mathematicians and learners grasp algebraic structures and its applications in real life scenarios.

This study seeks to explore the impact of GeoGebra software on teaching and learning maximum and minimum values of quadratic functions.

This chapter provides a Background to the study, Statement of the problem, Purpose of the study, Research questions, Hypotheses, Significance of the study, Delimitations and Limitations, and Organization of the study.

Background to the Study

Mathematics as a subject plays an important role in shaping problem solving, critical thinking, and analytic skills of students. Historically, mathematics education has faced persistent challenges. Globally, the academic performance of students in mathematics remains a top priority of educators, researchers, parents, teachers, other stakeholders, and Ghana as a whole cannot be exempted. The government of Ghana has implemented several policies over the years to

improve students' performance in mathematics in senior high schools. Policies such as Mathematics, Science, and Technology Education (MASTED), Science, Technology, Engineering, and Mathematics (STEM), and the National Science and Mathematics Quiz (NSMQ) have been implemented by the government to improve the academic performance of students in mathematics over the years. Despite all these efforts, students' performance over the years has been on a decline.

In recent years, educators have conducted research to address poor learner performance. Traditional teaching methods rely on static representations and manual calculations, which can hinder student engagement and understanding of complex concepts like quadratic functions. Governments and educational institutions globally have recognized the potential of technology to enhance learning outcomes, leading to initiatives aimed at integrating digital tools into the classroom (Johnson et al., 2016). The use of technological tools like the GeoGebra app has been explored as a means to bridge this gap, offering a dynamic, interactive platform that allows students to visualize mathematical concepts more clearly.

GeoGebra, a free, multi-platform software that combines geometry, algebra, and calculus, has been widely adopted in educational settings since its introduction. The app provides an interactive environment where students can manipulate mathematical objects, observe the effects of changes in real-time, and develop a better understanding of abstract concepts (Hohenwarter & Lavicza, 2015). Research has shown that using GeoGebra in teaching mathematics can significantly improve students' engagement and understanding of complex topics, such as quadratic functions (Saha & Ayub, 2015).

In Ghana, as in many other countries, there has been a growing interest in utilizing digital tools to enhance the learning experience in mathematics. The introduction of ICT into the school curriculum has paved way for adoption of tools like GeoGebra in classrooms (Amponsah, 2017). However, the effectiveness of these tools in improving student achievement, particularly in the context of quadratic functions, remains an area of ongoing research.

This study examines how the GeoGebra software influences high school students' ability to find the minimum and maximum values of quadratic functions. It aims to contribute to the discussion on technology in math education by providing evidence on the software's impact on student achievement.

Statement of the Problem

The teaching and learning of mathematics have been an issue of great concern to educators since mathematics as a subject is needed in everyday life. Mathematical concepts such as quadratic functions are a fundamental part of the senior high school curriculum. Mastery of these concepts will help learners to be functional in the society. These concepts are applied in fields such as economics, agriculture, engineering, among others. However, despite the importance of these functions, many students continue to struggle with them, hence they perform poorly (Jones & Smith, 2014).

There has been a global debate on integrating technology into education to improve teaching and learning outcomes. Studies have indicated that the teaching and learning of mathematics traditionally has contributed largely to students' low performance in Mathematics. According to Ottevanger et al. (2007), the traditional method of teaching mathematics does not help students

to gain deep understanding of Mathematical concepts. Hartsell, Herron, Fang and Rathod, (2009) criticized this method with the view that it does not adequately prepare students to excel in mathematics as it only focuses on teachers rushing to complete syllabus.

These global trends are mirrored in the African context, where mathematics education faces additional challenges, including limited access to quality educational resources and a lack of integration of technology in the classroom (Okereke, 2015). While some African countries have begun to incorporate technology into their curricula, the adoption of tools like GeoGebra remains limited, and their effectiveness in improving student achievement is not well documented. Research in this area has often focused on more general aspects of mathematics education, such as student attitudes towards mathematics or overall performance in national exams, rather than on specific technological interventions (Adu & Olabisi, 2017).

In Ghana, the West African Examinations Council (WAEC) has consistently reported poor performance of students in Mathematics. Performance of students in algebra and functions, remains a concern. Though the government and other stakeholders have been working on how to enhance mathematics education through means such as integrating ICT into the curriculum, there is still a significant gap in students' comprehension of quadratic functions (Amoako, 2018). From the foregoing discussion, it is obvious that students still have challenge in grasping the quadratic functions. Hence, this study seeks to use GeoGebra to teach the minimum and maximum values of quadratic functions to see its impact on students' performance.

Purpose of the Study

The purpose of this study is to investigate the effect of GeoGebra software on senior high students' achievement in determining the minimum and maximum values of quadratic functions. The purpose can be achieved by examining the difference in achievement scores of students receiving instruction through the use of GeoGebra software compared to the conventional method of teaching, and evaluating the effect of using GeoGebra software for teaching and learning of minimum and maximum values of quadratic function.

Objectives of the Study

The study was guided by the following objectives;

1. To ascertain whether there is a difference in the academic achievement between the pre-test and post-test scores of students receiving quadratic function instruction via the conventional way and the GeoGebra software.
2. To determine the effect of using the GeoGebra software on the achievement of senior high school students in determining the minimum and maximum values of a quadratic equation, compared to traditional teaching methods.

Research Question

The research question that underpinned the study is;

What is the effect of using the GeoGebra software on the achievement of SHS students in determining the minimum and maximum values of a quadratic function?

Hypothesis

The following null hypothesis was tested in the study:

H_0 : There is no statistically significant difference in the mathematics achievement test scores of students taught using the GeoGebra software and the same students taught using the conventional method.

Significance of the Study

This study looked into two senior high schools in the central region of Ghana. The study's primary focus was on students' abilities to determine the maximum and minimum values of a quadratic function. As a result, only one subject and one educational level could be used with the software. Although many other computer programs can be used as teaching tools for quadratic functions (Minimum and Maximum values), this study only examined the use of GeoGebra.

Delimitation

Only two senior high schools in Ghana's Central region were the subject of this investigation. The main focus of the study was how well students could identify a quadratic function's minimum and maximum values. As a result, the software's application was limited to a single subject and one educational level. This study solely looked at using GeoGebra, despite the fact that many other computer applications can be employed as teaching tools for quadratic functions (Minimum and Maximum values).

Limitations

There were restrictions on how widely the results could be applied because the sampling process relied on the deliberate selection of a sampling

unit. The research was conducted using a sample of senior high school students from the Central Region of Ghana. Students from two public senior high schools that followed the suggested scheduled academic cycle of the Ghana Education Service were present.

Organisation of the Study

There are five chapters in this work: Chapter One provides the study's background, problem statement, purpose of the study, research questions and hypothesis, significance of the study, delimitation, and limitations, as well as an organization of the study; Chapter Two reviews relevant literature; Chapter Three addresses methodology; Chapter Four presents' data analysis; and Chapter Five concludes with a summary, recommendations.

Chapter Summary

The general background offered in this chapter gave the research direction and focus. The first section of the chapter focused on research in mathematics education and how important it is to the field's advancement. The chapter also discussed interactive software as an innovative teaching and learning resource for quadratic function minimum and maximum values. The study's problem was then explained, highlighting the necessity of carrying out the investigation. The chapter goes on to outline the particular goals of the study as well as its aim. The research questions and hypothesis came next. The study's significance was then spoken about. The delimitation was presented together with a discussion of the limitations that were observed throughout the course of the investigation. The chapter summary came first, followed by the final portion, which is the study's organization.

CHAPTER TWO LITERATURE REVIEW

This Chapter explores various literature under the following themes: technology into teaching and learning, the use of technology in education in Ghana, the use of technology in Mathematics education, technology on students' mathematical performance, the teaching and learning of quadratic functions, some conceptual difficulties students encounter in learning functions, Chief examiners report on student's performance in Quadratic functions, and GeoGebra Software in Mathematics Education. The last sections provide the theoretical framework of the study and a summary of the chapter respectively.

Technology into teaching and learning

In today's educational landscape technology plays a role in shaping the learning environment for students and educators alike Bozkurt and Sharma (2020). The incorporation of technology tools and resources has improved the teaching methods and learning experiences in schools across the globe. One of the advantages of using technology in education is its ability to enrich the educational journey for students by offering interactive digital resources, like educational apps virtual simulations which are tailored to accommodate diverse learning preferences Greenhow and Askari (2017).

Another significant advantage of technology in education is its potential to improve the efficiency and effectiveness of teaching methods. Educators can utilize digital platforms to easily access and deliver educational content, track student progress, and provide personalized feedback Almusharraf and Khahro, (2020). Furthermore, the integration of technology in the classroom can also help address the diverse needs of students, as it allows for the implementation of differentiated instruction and adaptive learning strategies (Coy et al., 2018).

However, the integration of technology in education is not without its challenges. Arkorful and Abaidoo (2015) for example, identified funding as a significant obstacle to integrating technology in education in many developing countries. They found that educational institutions often struggle with inadequate budgets to invest in the necessary technological infrastructure, hardware, software, and ongoing maintenance and support. They also noted the need for more sustainable funding models and strategies. Additionally, they highlighted that in situations of limited resources, education administrators often have to prioritize other pressing needs over investments in educational technology, and the high upfront costs of purchasing devices, setting up connectivity, and acquiring digital content, especially for cash-strapped schools and districts. Arkorful and Abaidoo (2015) also emphasized the crucial need for consistent funding to cover the recurrent costs of hardware and software maintenance, upgrades, and capacity building, which is often overlooked. They concluded that sustainable funding mechanisms, public-private partnerships, and strategic resource allocation are essential to enable effective integration of technology in education, particularly in developing country contexts.

The use of Technology in Education in Ghana.

Ghana's education system is embracing technology with initiatives like the ICT in Education Policy (2008) and the Basic School Computerization policy (MoE, 2011). These programs aim to equip students with ICT skills, incorporate technology into teaching, and improve educational outcomes.

Studies by Natia Zonzongili (2012) highlight the positive impact of these efforts. Teachers are increasingly using ICT tools for lesson planning, delivery, and research. This can enhance learning by making it more interactive and

engaging. Additionally, technology allows for access to a wider range of educational resources, as shown in research by Kwei (2001). Students can utilize online tools for research, collaboration, and communication, fostering 21st-century skills.

However, challenges persist. Research by Boakye and Banini (2008) suggests limited technology use for academic purposes among students. Unequal access to infrastructure and resources remains a hurdle, with many schools lacking sufficient computers. Ghana's ICT Policy in education stresses the need for a balance in terms of ICT tools usage in schools. Furthermore, a gender gap exists, with Natia Zonzongili (2012) finding a disparity in male and female teachers' utilization of ICT tools.

Despite these obstacles, Ghana is making strides in integrating technology into education. The government's policies and initiatives provide a framework for progress. By addressing infrastructure limitations, promoting teacher training, and bridging the gender gap, Ghana can harness the full potential of technology to transform its educational landscape. One promising approach lies in utilizing low-cost mobile technologies. Research by Agyei and Danso (2013) suggests that mobile phones, with their high penetration rate in Ghana, can be effective tools for learning. Mobile apps and online platforms can deliver educational content, facilitate communication between teachers and students, and provide access to learning resources even in remote areas.

The use of Technology in Mathematics Education.

The landscape of mathematics education is undergoing a significant transformation with the integration of technology. This shift transcends simply replacing textbooks with tablets. Research by Hillmayr et al. (2020) highlights

the potential of technology-based interventions, like gamification and interactive software, to enhance students' achievement. Technology offers a plethora of benefits for both students and educators, fostering deeper understanding, increased engagement, and personalized learning experiences.

One of the most impactful ways technology aids mathematics education is by bringing abstract concepts to life. Through interactive visualizations and simulations, students can explore and manipulate mathematical ideas in a visually rich environment (Cai et al., 2020). This approach, as noted by Rashidov (2020), facilitates a stronger connection between abstract concepts and real-world applications, making mathematics more engaging and relevant. For instance, graphing calculators like Desmos allow students to experiment with functions and visualize their behaviour in real-time, fostering a deeper understanding of mathematical relationships.

The impact of technology extends beyond individual student achievement. Collaboration tools like video conferencing platforms enable interaction with experts and peers from around the world. Students can participate in discussions, share problem-solving strategies, and broaden their mathematical perspectives (Texthelp, 2020). This fosters a sense of community and global citizenship, reminding students of the vast applications of mathematics in various fields.

However, it is important to acknowledge that technology is a tool, not a silver bullet. Effective integration requires careful planning and pedagogical expertise. Teachers must ensure technology complements, rather than replaces, traditional instructional methods. Equitable access to technology and proper training for both educators and students are also paramount considerations.

When implemented thoughtfully, technology has the power to transform mathematics education into a dynamic, engaging, and empowering experience for all learners.

Technology on Student's Mathematics Performance.

There has been a lot of discussion and investigation about how technology affects pupils' performance in mathematics. While some studies have not discovered a statistically significant difference, others have found evidence that technology can improve children's math ability. As an illustration, research conducted in 2020 by Hillmayr et al. showed that students' performance in mathematics was enhanced by technology-based learning interventions such as gamification and interactive software. Additionally, students may self-correct and study at their own pace thanks to technology, which gives them quick feedback on their work.

As stated by Gómez-García et al. (2020), technology is not a magic bullet for improving mathematical performance. Technology can occasionally provide new difficulties for pupils, such as technical problems, interruptions, or insufficient instruction in its use. Furthermore, not every student can benefit from technology-based treatments, and they might need to be customized to fit each student's unique learning preferences and skill level. Furthermore, the way in which technology is applied may determine how effective technology-based therapies are (Birgin and Acar, 2020). For example, it might be more effective to use technology to enhance traditional classroom training rather than to replace it with in-person instruction (Zengin, 2018).

Technology can improve students' performance in mathematics, but the effectiveness of technology-based treatments depends on a number of variables,

including the technology's quality, the instructional strategies used, and the particular needs and skills of the students.

Teaching and Learning of Quadratic Functions.

A study of the quadratic function is essential for pupils to succeed in more sophisticated mathematical concepts. It is a fundamental topic in mathematics. When it comes to their minimum and maximum values, quadratic functions which have the equation $y = ax^2 + bx + c$, display special qualities. The mathematical foundations and real-world applications of these important points have been studied in this field of study. The existence of a vertex, or the function's minimum or maximum value, is a basic feature of quadratic functions. The vertex's location can be ascertained by calculating the values of the coefficients a , b , and c . Specifically, the formula $x = -\frac{b}{2a}$ yields

$2a$

the vertex's x -coordinate, which can then be substituted into the original equation to obtain the y -coordinate. Understanding the vertex is crucial for tasks such as optimization problems and graphing quadratic functions

The use of diagrams and visual representations can be an effective approach in teaching quadratic functions. Researchers have emphasized the importance of using visual aids to enhance students' understanding of the properties and characteristics of quadratic functions.

Bossé and Nandakumar (2005) looked into the effects of teaching quadratic functions with graphing calculators. They discovered that students' conceptual grasp of the key components of quadratic functions, such as the vertex, axis of symmetry, and graph transformations, improved when they used graphing calculators.

Additionally, Borba and Confrey (1996) discussed the importance of incorporating real-world applications and contextualized problems in the teaching of quadratic functions. They argued that this approach can help students see the relevance of the topic and motivate them to engage more deeply with the material.

The teaching of quadratic functions is crucial for developing students' mathematical reasoning and problem-solving skills. Researchers have emphasized the importance of incorporating multiple approaches to the learning of quadratic functions.

Kemp and Kissane (2010) highlighted the value of using graphical, symbolic, and contextual representations to enhance students' conceptual understanding of quadratic functions. They suggested that by engaging students in modelling real-world situations with quadratic functions, educators can foster the development of deductive reasoning and problem-solving abilities. Similarly, Stump (2001) argued that the teaching of quadratic functions should emphasize the connection between the algebraic, numerical, and graphical representations. By encouraging students to explore these multiple representations, educators can help them develop a deeper understanding of the underlying concepts and the ability to apply quadratic functions in various contexts.

Using games is also an effective way to create engaging activities for teaching and learning quadratic functions. Researchers have demonstrated the benefits of this approach. For example, a study by Smith, Ivanov, and Lee (2021) investigated how a teacher can use a game-based approach to help students develop a deeper understanding of quadratic functions. In their study,

the researchers worked with 47 high school students in an algebra class. The students participated in a game called "Parabola Pursuit" that focused on the properties of quadratic functions. The goal of the game was for students to identify key features of parabolas, such as the vertex, axis of symmetry, and rate of change. The researchers first gave the students a pre-assessment to gauge their prior knowledge of quadratic functions. They then designed and implemented the game-based instructional sequence over the course of three weeks. During the game, students had to analyze the characteristics of different parabolas and compete to correctly identify them. The findings revealed that after playing the game, students showed significant improvements in their ability to recognize and analyze the properties of quadratic functions. The game allowed them to actively engage with the concepts, rather than just passively receiving information. The researchers concluded that incorporating well-designed games into the curriculum can be a powerful way to enhance students' learning and understanding of quadratic functions.

In the context of learning quadratic functions, students' thought processes are also crucial in demonstrating their understanding. A study by Zaslavsky (2005) explored the strategies and misconceptions that students exhibit when working with quadratic functions. The researchers investigated a sample of 85 high school students in Israel, focusing on their approaches to solving problems related to quadratic functions.

The study found that students often relied on naive or intuitive methods when dealing with quadratic functions, which led to various misconceptions. For instance, some students struggled to grasp the relationship between the coefficient of the quadratic term and the shape of the parabolic graph. Others

had difficulty interpreting the meaning of the vertex and its relationship to the solutions of the quadratic equation.

Additionally, the researchers observed that students' visual perceptions of the graphical representations of quadratic functions influenced their understanding. Students tended to focus more on the overall shape of the parabolic graph rather than the underlying mathematical properties, which sometimes resulted in incorrect conclusions about the behaviour of quadratic functions. The study also highlighted the importance of providing students with varied and engaging activities to enhance their learning. The researchers found that students demonstrated better understanding when they were given opportunities to explore quadratic functions through hands-on activities, such as using graphing calculators or interactive simulations.

One relevant study was conducted by Smithson and Lester (2011), who investigated the factors that influence students' understanding of the conditions required for a function to have a minimum or maximum value. They found that many students struggle to correctly identify the critical points of a function and distinguish between minimum and maximum values of a function. This highlights the need for more pedagogical approaches that help students develop a deeper conceptual grasp of extremum problems.

Another study by Varberg and Purcell (2007) explored the use of graphical representations to enhance students' abilities to determine the minimum and maximum values of a function. They found that the visual interpretation of a curve's behaviour was often more intuitive for students than purely analytical methods, suggesting that a balanced approach incorporating both graphical and algebraic techniques may be beneficial. Further research has

delved into the conditions that determine whether a quadratic function has a minimum or maximum value. When the coefficient is positive, the function will have a minimum value, while a negative value of a results in a maximum value (Stewart, 2016). This distinction is essential for applications where finding the optimal solution is paramount, such as in engineering design or economic decision-making.

Overall, the minimum and maximum value of a curve is a fundamental concept in mathematical analysis with important applications in various fields. Further research is needed to better understand how students engage with and apply this knowledge, in order to develop effective instructional strategies that foster a deeper appreciation for optimization problems and the properties of extrema.

Conceptual Difficulties Students Encounter in Learning Functions

One mathematical topic that students find difficult to master is quadratic functions because of inefficient teaching methods. As mentioned by Benning and Agyei (2016), Dubinsky and Harel, (1992), Eraslan (2005). Several studies have documented the quadratic function learning challenges of these children.

Benning and Agyei (2016) state that Eraslan's (2005) research revealed four cognitive barriers that students encounter when learning quadratic functions. The inability to draw mathematical connections between algebraic and graphical aspects of concepts, the need to familiarize oneself with a foreign concept, the disequilibrium between algebraic and graphical thinking, and the image of the quadratic formula or absolute value function are some examples of these cognitive barriers. Students are not able to draw and explore mathematical

links between the pictorial and algebraic parts of the quadratic functions, according to Eraslan.

Benning and Agyei (2016) also mentioned that Clement (2001) found that students may find the most difficulty with concepts, images, and the link between them when they are converted from graphical to algebraic form. Several studies have shown that both the algebraic and graphical techniques of representing images and pre-images only partially capture the concept and representation of images (Dubinsky and Harel, 1992; Eraslan, 2005; Looney, 2004; Maharaj, 2008). Although graphing functions is an essential component of comprehending quadratic functions, research by Dubinsky and Harel (1992); Eraslan (2005); Looney (2004); and Maharaj (2008) regularly demonstrates that students find sketching quadratic function graphs difficult and confusing. Eraslan (2005) so pointed out that students should be receptive to related concepts like turning points while teaching quadratic functions.

Chief Examiners Report on Student's Performance in Quadratic Functions.

The Chief Examiner's Reports for the West African Senior School Certificate Examination (WASSCE) (West African Examinations Council, WAEC), 2007; 2008) brought attention to cases of subpar student performance on national examinations. Students made progress in areas such as binary operations, binomial theorem, and series, however they had difficulty factorizing quadratic equations of the form $ax^2 + bx + c$ where $a \neq 0$. This was shown in the 2007 Chief Examiners' Report. Furthermore, a lot of students found it challenging to solve equations utilizing the idea of equal roots in quadratic functions. For instance, pupils encountered difficulties when

attempting to answer questions such as: The quadratic equation's roots : $x^2 - 2(3k + 1)x + 7$ where k is a constant, are equal. Find the values of k (WAEC, 2007, p. 60).

The report highlighted that several students faced challenges in solving this problem. The 2008 report also emphasized that students struggled with conceptual issues when sketching quadratic graphs (WAEC, 2008).

The discussion above implies that the way the subject is organized and presented may be contributing to students' difficulties in comprehending quadratic functions. For this reason, it's critical that math teachers concentrate on teaching functions in a way that makes sense to their students. Monaghan (2005) emphasized the relationship between numerical, symbolic, and graphical representations to demonstrate functions using technological software. This study examined the effects of technology-enhanced training on senior high school students' performance in comprehending the minimum and maximum values of quadratic functions. The GeoGebra application software was utilized as a teaching tool.

GeoGebra Software in Mathematics Education.

GeoGebra, the dynamic mathematics software, was invented by Markus Hohenwarter, an Austrian mathematician and computer scientist. In 2001, while a Ph.D. student, Hohenwarter developed GeoGebra as a tool to integrate geometry, algebra, and calculus into a single interactive platform. GeoGebra's open-source nature and user-friendly interface have made it a widely adopted educational software in classrooms worldwide.

For students at various educational levels, GeoGebra is an adaptable mathematics program. Within a single platform, it combines calculus, statistics,

graphing, geometry, algebra, and spreadsheets. The online platform offered by GeoGebra also boasts an extensive library of more than a million free teaching resources. Real-time monitoring of student progress is made possible by the easy sharing of these resources via the collaborative GeoGebra Classroom.

As the industry leader in dynamic mathematics software, GeoGebra promotes breakthroughs in teaching and learning and supports STEM (science, technology, engineering, and mathematics) education across the globe. Numerous educational websites across the globe, ranging from basic demonstrations to comprehensive online assessment systems, rely on GeoGebra's math engine. The GeoGebra software interface is shown in the figure below.

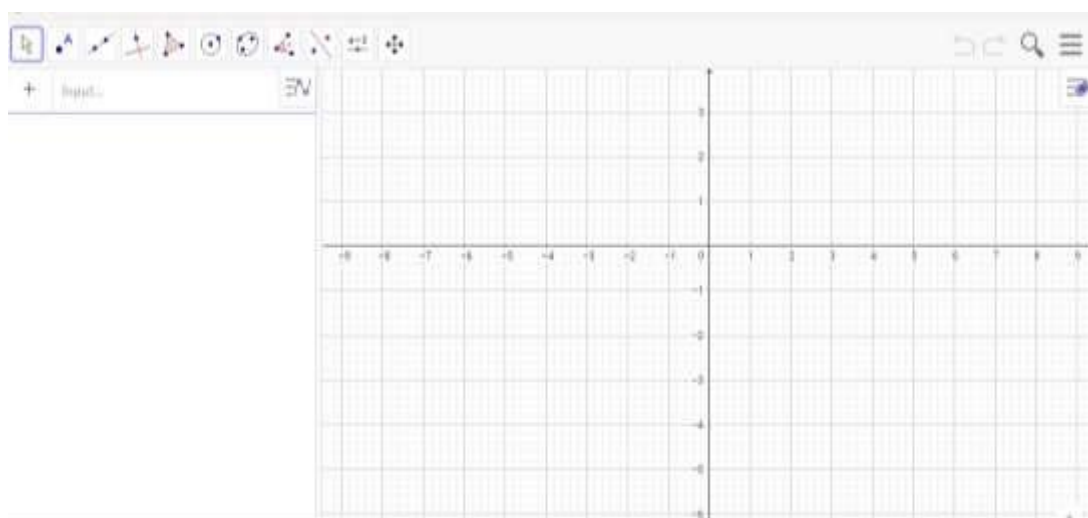


Figure 1: GeoGebra working area

Source: GeoGebra software

GeoGebra can be used in the mathematics classroom to facilitate interactive learning of quadratic principles. After explaining the topic to be taught, the teacher uses GeoGebra tools to visually illustrate or animate the notion to help the students grasp it better.

Using GeoGebra in the classroom helps to facilitate dynamic and exploratory learning of mathematical concepts. Unlike static representations in textbooks, GeoGebra allows students to manipulate parameters of quadratic functions and other mathematical objects in real-time, helping them develop a deeper, more intuitive understanding Beigie,(2011). For example, students can investigate the properties of quadratic functions, observe how changes in one parameter of a function affect its graph, or experiment with the effects of varying parameters in mathematical models Trocki and Hollebrands, (2018).

Furner and Marinas (2021) conducted a study with high school students, examining the impact of GeoGebra on their understanding of the graphical representation of quadratic functions. The researchers found that students who used GeoGebra demonstrated a deeper understanding of the relationship between the coefficients of a quadratic function and the shape of its graph, as well as improved problem-solving skills. Similarly, Dikovic (2019) explored the integration of GeoGebra in a college-level algebra course, focusing on the teaching of quadratic functions. The study revealed that students who used GeoGebra were better able to visualize and interpret the effects of changes in the function parameters on the graph, leading to a more comprehensive understanding of quadratic functions.

The efficiency of GeoGebra in teaching middle school children about quadratic functions was investigated in research by (Abar and Barcelos 2020). By merging quantitative and qualitative data, the researchers used a mixedmethods strategy. As evidenced by higher evaluation scores and more thorough explanations of mathematical ideas, the usage of GeoGebra enhanced students' understanding of quadratic functions.

A study conducted by Zengin (2017) examined how using GeoGebra affected students' understanding of quadratic functions and their capacity to recognize minimum and maximum values. Sixty high school pupils were involved in the study; they were split into experimental and control groups. While the control group received conventional training, the experimental group used GeoGebra to learn about quadratic functions. The experimental group outperformed the control group in identifying the lowest and maximum values of quadratic functions, according to the data, indicating that GeoGebra can help students better understand this idea conceptually. Similarly, Dikovic (2009) looked at how GeoGebra could be used to educate students how to graph quadratic functions and find their lowest and maximum values. 50 university students participated in the study, which discovered that using GeoGebra in the classroom improved students' comprehension of the properties of quadratic functions, including as their minimum and maximum values.

Scholars have also examined the unique characteristics of GeoGebra that enhance its efficacy in instructing students on the minimum and maximum values of quadratic functions. The dynamic features of GeoGebra, which let students adjust a quadratic function's parameters and see how the graph changes in response, as well as the lowest and maximum value, were emphasized by Hohenwarter and Preiner (2007). This dynamic visualization can help students develop a deeper understanding of the relationship between the function's coefficients and its minimum or maximum value.

Furthermore, Tatar and Zengin (2016) explored the use of GeoGebra's slider tool to investigate the effects of changing the coefficients of a quadratic function on its minimum or maximum value. Their study found that the use of

the slider tool facilitated students' exploration and understanding of the relationship between the function's coefficients and its extremum points.

Theoretical Framework for the Study

The Diffusion of Innovation Theory by Rogers (2010), which holds that a new technology's perceived qualities affect its acceptance and diffusion (in this case, the GeoGebra program), serves as the foundation for this study. The idea contends that factors such as GeoGebra's relative benefit, compatibility, complexity, testing, and observability will determine how widely it is adopted and how well it is used to teach and understand quadratic functions.

The degree to which GeoGebra provides advantages over conventional teaching approaches is known as relative advantage. The degree to which the program is compatible with current educational methods and philosophies is evaluated. Complexity assesses how simple or challenging it is for educators and learners to comprehend and utilize GeoGebra. Testing gauges how much a teacher can use GeoGebra in a small-scale trial before deciding to fully use it. The visibility of GeoGebra's influence on teaching and learning outcomes is referred to as observability.

This study intends to explore the spread of GeoGebra in teaching quadratic functions by looking at these characteristics, exposing its potential to improve student comprehension and engagement. By understanding the elements that impact the uptake and efficient application of GeoGebra, this study can provide insights into tactics for encouraging the fruitful incorporation of technology into mathematics instruction.

Chapter Summary

This chapter offers a survey of the literature that emphasizes numerous investigations into the application of technology—more especially, GeoGebra Software—in the instruction and acquisition of quadratic functions. Although a lot of study has been done on quadratic functions in general, not much of it has been done on using GeoGebra software to comprehend quadratic function extrema. Prior research has focused on how well students can recognize mathematical objects and improve their deductive and algebraic reasoning abilities when addressing quadratic function problems using computer programs. Nevertheless, there is a deficiency in the way that extrema of quadratic functions are addressed and how GeoGebra software is used to improve these ideas in teaching and learning. This study is notable for its distinct focus on Ghanaian Senior High School students and their knowledge of quadratic function extrema.

CHAPTER THREE RESEARCH METHODS

This chapter present the research design, study area, population sample, sampling technique, data collection instrument(s), reliability and validity of the research instruments, data collection procedures and the data processing and analysis.

Research Design

The cause-and-effect relationship between an independent and dependent variable was examined using a quasi-experimental method in this study. When evaluating the effects of an intervention on a particular population without assigning participants at random, researchers employ quasiexperiments (Creswell, 2014). The study's approach made use of a control group of students

who were taught quadratic functions using GeoGebra software and conventional ways.

The researcher was able to examine cause and effect in order to determine the effects of incorporating GeoGebra software into teaching and learning on students' performance.

This study examines the effects of teaching senior high school pupils the minimum and maximum values of quadratic functions using the GeoGebra program. The researcher intends to give the pupils the lesson in order to accomplish this. A comparison is made between the pre- and post-GeoGebra performance of the same group of students. First, a standard teaching strategy is used with an identical group of pupils, utilizing a marker board and markers. After that, the GeoGebra software is used to instruct them.

Study Area

The University of Cape Coast and a number of other prestigious secondary institutions are located in Ghana's Central Region, which is also a major center for education in the nation. Situated in Ghana's South-Western Center, the region is bounded to the north by the Ashanti and Eastern regions, to the west by the Western region, to the east by the Greater Accra region, and to the south by the Gulf of Guinea. Cape Coast, the region's central city, is renowned for its rich history, picturesque setting, and historical significance. The city enjoys a tropical savannah climate with consistent temperatures yearround, influenced by its proximity to the Atlantic Ocean, which enhances its landscape with lush greenery and stunning ocean views.

The Central Region of Ghana is divided into various districts, each with its own unique characteristics and attractions. Some of the notable districts in the Central Region include Cape Coast Metropolis, Agona West Municipal, Gomoa West District, and Awutu Senya East Municipal. These districts encompass diverse communities and landscapes, ranging from urban centres to rural areas. Cape Coast, the capital city of the Central Region, serves as a key cultural and educational centre, while the surrounding districts also contribute to the region's economic and social progress in diverse ways. Each district has its own local government structure and helps shape the overall identity and attractiveness of the Central Region. The Cape Coast Metropolis district, a crucial center for education and culture in Ghana's Central Region, is the primary focus of the research. The Cape Coast Metropolis neighborhood, wellknown for its historical significance and for housing prestigious universities including Cape Coast Technical University and University of Cape Coast, is an important driver of regional growth. The district's standing as a hub for academic success in the area has been largely attributed to its strong emphasis on education. The Cape Coast Metropolis district is composed of both rural and urban areas geographically. The district encompasses coastal areas with beautiful beaches and historic landmarks such as Cape Coast Castle, which holds great historical significance. The district's diverse geography makes it a notable destination for both educational and recreational purposes, attracting students, scholars, and tourists alike. Two Senior High Schools were purposefully selected from this district for the study.

Population

The study population includes all Senior High School students in the Central Region of Ghana, totalling 56 schools, both public and private. The specific focus is on Form Two students in the region. The study was conducted at School A and School B, where students received instruction on quadratic functions through both traditional methods and the GeoGebra software.

Sample and Sampling Technique.

The Central Region of Ghana, which consists of 22 districts, is where the target population was found for the sample. For the study, two districts were selected at random: Mfantseman Municipal District and Cape Coast Metropolitan District. Purposive sampling was used to determine which two Senior High Schools have adequate ICT facilities. Form Two classes at these schools were purposefully chosen since the researcher's topic—quadratic functions—fits into the Form Two curriculum as per the Senior High School Mathematics Curriculum. For the study, one Form Two class from each of the two schools—A and B—was selected at random. Both the traditional method and GeoGebra are being used in the instruction of the chosen Form Two classes in both schools.

Data Collection Instrument(s)

We employ three instruments to examine the relationship between the independent variable (achievement test) and the dependent variable (instruction using GeoGebra and traditional teaching methods): a pre-test, a post-test, and a questionnaire. Prior to the interventions, students' skill with quadratic functions is assessed using a pre-test. The post-test assesses how the treatments have

affected students' learning. The pre-test items are based on the Senior High School Mathematics curriculum's quadratic functions topic, while the post-test items focus on the treatment topic of minimum and maximum values. A questionnaire was given to students after the intervention to gather feedback.

Pre-test and Post-test Instruments

There is one theoretical question on quadratic functions in both the pre- and post-tests that students need to respond to. The purpose of the pre-test is to evaluate students' readiness for learning about the minimum and maximum values of quadratic functions by gauging their prior knowledge and comprehension.

Following the intervention, the post-test assesses students' comprehension of the ideas and connections in Maximum and Minimum. The quantity and content of the post-test are the same for the same set of pupils.

Questionnaire

Students complete the research questionnaire to gather data on their attitudes, motivations and acceptance of using GeoGebra in mathematics education following the GeoGebra intervention. The questionnaire is filled out by the study participants. There are three parts to the survey. The section A comprises four items on students' demographics, section B comprises seven items on how GeoGebra has influenced students understanding of quadratic functions, section C comprises eight items on students' opinion on using GeoGebra for learning Mathematics.

Reliability and Validity of Research Instruments

The researchers guarantee the reliability and validity of the instruments through pre-testing. This process helps detect any errors in the test items and confirms their relevance to the study's objectives. Additionally, two experienced mathematics teachers are tasked with scoring the students' work in both the pretest and post-test to enhance reliability. The pre-test and post-test ensure content validity for the current study. Content validity focuses on how well a measurement tool, such as a test, survey, or experiment, accurately reflects the specific construct it's designed to measure (McGuire et al., 2018).

Data Collection Procedures

The researchers will build a strong relationship with students in the selected schools for the study. They have obtained approval from the Department of Mathematics and ICT Education at the University of Cape Coast to conduct the research. The researchers obtain consent from the Department and then present a letter of introduction to the Heads of the selected schools before the data collection exercise.

Administration of Pre-test Instrument

The same set of students take the pre-test a few days prior to the intervention. On quadratic functions, there is an open-ended question. Data is saved on a computer once pre-test papers have been marked.

Instruction on quadratic function

During the intervention, participants are taught about Minimum and Maximum values over the course of a day. The identical group currently learning Maximum and Minimum values in a GeoGebra-supported

environment. For the same group, the researcher doubles as an instructor. The researcher created the lesson plans and materials before the intervention period and delivered them to field teachers for their approval. The researcher evaluates every lesson plan and exercise before it is put into practice.

For an entire day, the same group is being instructed in the traditional way. Instructional lessons are being developed by the researcher and teachers from both School A and School B. In this research, the conventional teaching method involves the teacher using a marker board and marker to guide students through each step of Minimum and Maximum values in Schools A and B. The researcher takes on the role of the teacher for this group.

The intervention included teaching:

1. The Minimum value of a quadratic graph 2.

The Maximum value of a quadratic graph.

Using Conventional Teaching.

The researcher begins the lesson by reviewing relevant previous knowledge. Students are asked to fill a table for a given equation. Researcher introduces the Minimum and Maximum values to the students.

The researcher examines the concept of maximum and minimum curves and illustrates examples on the marker board to demonstrate different ways of representing these curves.

Using the GeoGebra Software

The lesson starts with the researcher introducing the fundamental tools of the GeoGebra software to the students, taking approximately five minutes for the introduction. After the introduction to the basic tools in GeoGebra, the

Figure 2 presents a diagram of the minimum values of the quadratic function. The minimum value is also called the lowest value. GeoGebra software helps students easily visualize the concept and ideas of a Minimum value. Figure 3 shows a diagram of the Maximum values of the quadratic function. The maximum value is also called the greater value.



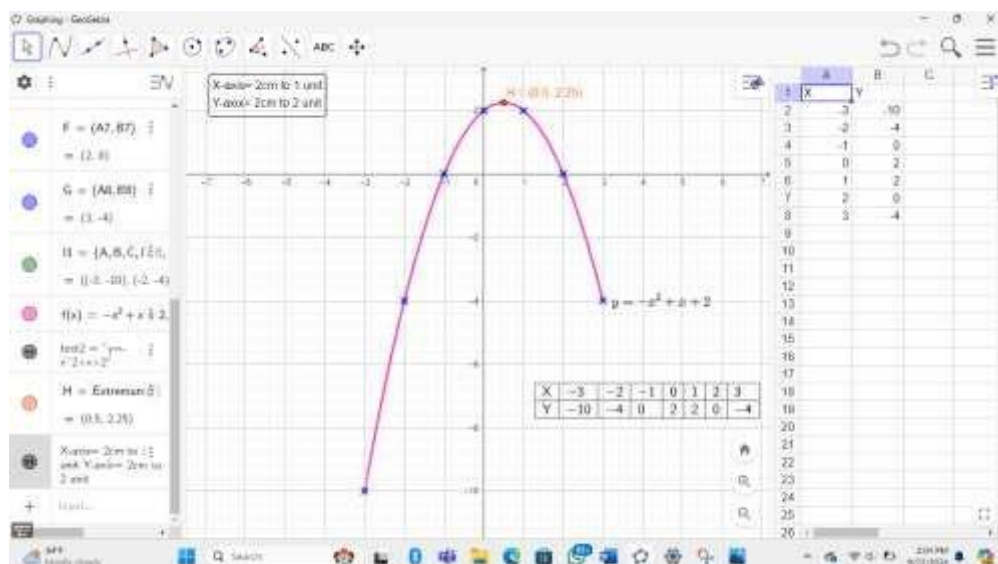


Figure 3: Maximum or Greatest value

Administering Post-test and Research Questionnaire

After each session, participants are given post-tests, one after the traditional teaching method and another after the GeoGebra intervention. The tests are graded, and the scores are entered into the computer as quantitative data.

The researchers compare the results of the conventional teaching test scores and the GeoGebra intervention test scores after administering both posttests. This comparison allows them to look into the current effect of the GeoGebra software on student learning compared to the conventional teaching approach.

A research questionnaire is distributed to collect students' feedback on the effectiveness of the software in teaching quadratic functions. The collected data is then organized for analysis and interpretation of the findings.

Data Processing and Analysis

Data analysis requires the use of various techniques, including

descriptive and inferential statistics.

Descriptive Statistics

To characterize and summarize the data in an understandable way, descriptive statistics are used. These typically include measures such as means, medians, standard deviations, and frequency distributions, providing an overview of the participants' performance on the post-tests.

The data in this study are described using the mean and standard deviation. The pre- and post-test items, as well as the questionnaire, are analyzed using percentages.

Inferential Statistics

Inferential statistics are applied to draw conclusions and determine the statistical significance of the findings. This involves hypothesis testing, regression analysis, or other inferential methods to assess whether the observed differences between teaching with the conventional method and teaching with the GeoGebra intervention are statistically meaningful. In this study, the researchers use T-test statistical technique to assess if there is a notable disparity in achievement scores among students studying Mathematics with the GeoGebra software versus the traditional approach.

By combining these complementary analytical approaches, the researchers are able to provide a robust and comprehensive analysis of the data collected throughout the study.

Chapter Summary

This study, which focuses on two schools in Ghana's Central Region, uses a quasi-experimental methodology. Simple random sampling and purposive sampling are used in the sample selection process to choose two Form 2 classes

from various local schools. Pre- and post-tests, as well as surveys, are some of the data collection techniques. Statistical techniques for both inferential and descriptive analysis are used in the data analysis.

CHAPTER FOUR FINDINGS AND DISCUSSIONS

In keeping with the goals and purposes of the study, the chapter offers the findings and examines the research. The goal of the study was to determine how senior high school students' use of the GeoGebra program affected their capacity to determine the minimum and maximum values of quadratic functions. The study's goals were looked at, and the findings were evaluated in light of the research question and null hypothesis. The data were presented and examined using tables, figures, descriptive, and inferential statistics. The chapter was concluded with a summary.

Preliminary Analysis.

The study sample comprises two senior high schools named School A and School B. A form two class was selected for the study at each school. 32 students comprising only boys were used in school A (a single sex school) whereas school B was sampled 43 students comprising 25 boys and 18 girls. In all, 75 students from the two schools from the districts were the participants for the study. Pre-test and post-test were both administered to the students before and after instructing them using GeoGebra. Following the post-test, a survey was presented to obtain information on students' attitude, motivation and acceptance toward teaching and learning of quadratic function using GeoGebra software.

Findings from Pre-Test

The pre-test was administered prior to providing intervention to assess students' ability to determine the minimum and maximum values of quadratic functions following conventional teaching methods. The test was scored out of 10 (see Appendix A). A score of 5 out of 10 was considered average, while a

score below 5 was considered below average. Scoring above 5 out of 10 was graded as an above-average performance. As shown in Table 1, percentages were used to analyze the pre-test results.

Table 1: Descriptive statistic findings for students' pre-test

Scores	Frequency	Percentage
0	3	4.1
1	5	6.7
2	4	5.3
3	10	13.3
4	14	18.7
5	28	37.3
6	7	9.3
7	4	5.3
Total	75	100

Source: Filed Work, (2024)

Table 1 represents the pre-test scores collected from the respondents of the study. The table presents a distribution of scores along with their corresponding frequencies and percentages. The outcome of the test reveals a broad distribution of performance. The most frequent score is 5, with a frequency of 28, making up 37.3% of the total. According to the study's findings, 28 out of 75 pupils' achievement levels are within the average range. The second most common score is 4, with a frequency of 14, accounting for 18.7% of the data.

However, a total of 36 students achieved scores that fell below the average, which accounts for 48.1% of the group. In contrast, 11 students managed to score above the average, representing 51.9% of the entire cohort. This distribution highlights the performance variation within the student body.

Findings from Post-Test

After teaching them to use the GeoGebra program, the post-test was given to gauge the students' comprehension of finding the minimum and maximum values of quadratic functions. Additionally, the post-test had a score of out of ten (see Appendix B). Similar to the pre-test, the post-test was scored, with a score of five out of ten representing an average performance and a score of less than five being below average. A performance rating of more than five out of ten was considered above average. Percentages were used to analyze the post-test data, as indicated in Table 2.

Table 2: Descriptive statistic result for students' post-test

Scores	Frequency+	Percentage
3	2	2.7
4	5	6.7
5	8	10.7
6	4	5.3
7	11	14.7
8	15	20.0
9	22	29.3
10	8	10.6
Total	75	100

Source: Filed Work, (2024)

Table 2 represents the scores of the post-test collected from the respondents of the study. It was observed from the post-test results that, 60 out of the 75 students obtained scores ranging from 6 to 10, indicating performance above average. Thus, 79.9% of the students performed above average. The findings of the test also indicated that 8 students performed averagely with a score of 5 out of 10, representing 10.7%. However, 7 students, representing

9.4%, had scores below 5 in the post-test, falling below average.

Difference in Academic Performance between Pre-test and Post-test Results of the identical sets of students

In other to address the study's initial goal the pre and post-test results were analysed together. The post and pre-test results were discussed to test the study's hypothesis using descriptive and inferential statistics. The two test scores' means and standard deviations were compared, and the t-test statistic was used to determine whether there was a statistically significant difference between them.

The hypothesis that guided the study is: H_0 : There is no statistically significant difference in the mathematics achievement test scores of students taught using the GeoGebra software and the same students taught using the conventional method. This was tested to examine the cause and effect of the GeoGebra instruction on the students' pre-test and post-test.

Table 3: Pre- and Post-Test Descriptive Statistics for Students' Scores

Group	N	Mean	Std. Deviation
Pre-test	75	4.11	1.682
Post-test	75	7.53	1.890

Source: Filed Work, (2024)

Table 3 presents a comparison of the pre- and post-test scores, along with the number of participants (N), mean (average), and standard deviation. According to the above table, the post-test mean and standard deviation are 1.890 and 7.53, respectively, while the pre-test mean is 4.11 and the standard deviation is 1.682. The increase in the mean indicates that, on average, participants' scores improved after the GeoGebra intervention. The difference in standard deviation from 1.682 to 1.890 suggests that while most participants improved, there was a wider range of performance in the post-test scores as

compared to the pre-test scores. This means that the extent of improvement varied slightly more among participants after the GeoGebra intervention.

In summary, table 3's statistics show that participants' performance improved between the pre- and post-tests. This result suggests that, presumably, the intervention was successful in improving the respondents' overall performance; nevertheless, more analysis using a paired sample t-test may be necessary to determine whether there has been a significant increase.

Analysis of Students' Pre-Test and Post-Test Scores Using Inferential

Statistics.

Table 4: Paired Samples t-test of students' pre-test and post-test scores

	t	df.	Sig.(2 tailed)
Pair 1 Post-test- Pre-test	11.610	74	.000

Source: Filed Work, (2024)

There is a statistically significant difference between the pre-test and post-test scores, as shown by the paired t-test results. this is due to the fact that the alpha value of.05. is less than the Sig (2-tailed) of.000. Additionally, table 1.5a indicates that the pre-test mean score was 4.11 and the post-test mean score was 7.53. This indicates a significant rise in the post-test scores, indicating that students performed better in the post-test than in the pre-test. The statistical significance of the difference between the pre- and post-tests is indicated by the large t-value (11.610) and the p-value (.000). This result defies the null hypothesis, which holds that there is no statistically significant difference between the math achievement test results of pupils taught using the traditional technique and the GeoGebra program.

Discussion related to the null hypothesis (H_0)

The research's conclusions clearly refute the null hypothesis by showing that there is a statistically significant difference between the study's pre- and post-test scores. This indicates that there is a noteworthy distinction between the achievement scores of pupils learning mathematics through the use of the GeoGebra software and those who do not. The research results are consistent with those of Brown et al. (2019), who discovered that students utilizing GeoGebra had a significant increase in test scores, suggesting that the incorporation of technology in math teaching can result in improved outcomes. The null hypothesis of the study, which states that, there is no statistically significant difference in the mathematics achievement test scores of students taught using the GeoGebra software and the same students taught using the conventional method., is rejected. These findings are consistent with the study that Zengin, Furkan, and Kutluca (2012) carried out. According to their research, there was a notable distinction in the mathematical proficiency of pupils who used GeoGebra in their classes compared to those who did not.

This conclusion is further supported by the work of Shadaan and Leong in (2013), who also observed that students taught using GeoGebra showed statistically significant improvements in their mathematics performance compared to their peers who were instructed through conventional teaching methods. These findings highlight the positive impact of integrating GeoGebra into mathematics education, suggesting that the software can be a valuable tool in enhancing students understanding and achievement in this subject area.

To summarize, the paired t-test provides compelling evidence of a noteworthy distinction between the two sets of data. Thus, there is a statistically significant difference in the mathematics achievement test results between students taught using the GeoGebra program and the same students taught using the conventional technique, at 95% confidence interval, with degree of freedom 74.

These findings suggest that GeoGebra can be a valuable tool for enhancing students' understanding and proficiency in determining the maximum and minimum values of quadratic functions. These findings contribute to the broader discourse on the role of technological tools in teaching and learning to help overcome challenges in mathematics education, reinforcing the importance of technological tools like GeoGebra, which can help students achieve educational success.

Students' Attitude, Motivation and Acceptance towards GeoGebra Instruction

This study also delved into assessing students' attitudes, motivation and acceptance of the GeoGebra instruction in determining the minimum and maximum values of quadratic function. A questionnaire comprising 15 items was used to obtain information on the attitudes of students, their motivation and acceptance of the GeoGebra instruction to answer the research question of the study. Students answered the statements using a 4-point Likert scale.: Key: SA = Strongly Agree, A = Agree, D= Disagree, SD = Strongly Disagree. **The Research question that guided the study is:** What is the effect of using the GeoGebra software on the achievement of senior high school students in determining the minimum and maximum values of a quadratic functions?

Table :5 Students' attitude, motivation and acceptance of the GeoGebra instruction.

Statement	SA %	A %	D %	SD %
After using GeoGebra I am confident in determining the minimum and maximum values of quadratic functions.	57.3	36.0	5.3	1.3
GeoGebra has helped improve my understanding of the concept of maximum and minimum values of a quadratic function.	60.0	29.3	8.0	2.7
GeoGebra has improved my ability to solve problems related to maximum and minimum values of a quadratic function.	53.3	37.3	6.7	2.7
The GeoGebra app has helped me visualize the relationship between the coefficients of a quadratic function and its minimum or maximum value.	40.0	45.3	12.0	2.7
I can use the GeoGebra app to verify the solutions of quadratic equations.	20.0	34.7	33.3	12.0
I can use the GeoGebra app to investigate how changes in the coefficients of a quadratic function affect its minimum or maximum value.	14.7	26.7	37.3	21.3
The GeoGebra app has helped me better understand the real-world applications of quadratic functions.	42.7	40.1	14.7	2.7
I enjoy using the GeoGebra app for my mathematics learning.	50.7	38.7	9.3	1.3
The GeoGebra app has made me more interested in learning mathematics.	53.3	32.0	10.7	4.0
I find the GeoGebra app to be a useful tool for learning mathematics.	58.7	36.0	1.3	4.0
The GeoGebra app has made me more interested in learning mathematics.	53.3	32.0	10.7	4.0
The GeoGebra app has helped me learn mathematics at my own pace.	42.7	46.7	5.3	5.3
I would recommend the use of the GeoGebra app to other students.	54.7	37.3	6.7	1.3
The GeoGebra app should be integrated into the mathematics curriculum.	56.0	38.7	5.3	0.0

The GeoGebra app should be integrated into the mathematics curriculum.	56.0	38.7	5.3	0.0
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Source: Filed Work, (2024)

A substantial number of students gained confidence in figuring out the minimum and maximum values of quadratic functions following the GeoGebra intervention, according to Table 1.5's analysis of students' attitude, motivation, and acceptance of the GeoGebra instruction. Of those surveyed, 57.3% strongly agreed with this statement. Likewise, a significant number of students (60.0%) agree that GeoGebra has been helpful in helping them better understand the idea of the maximum and lowest values of a quadratic function. This was further demonstrated by the high degree of agreement, with 58.7% of students saying that the GeoGebra program is a helpful resource for learning mathematics and 53.3% of students strongly agreeing that it has improved their capacity to solve problems involving the maximum and minimum values of a quadratic function. 40% of the students confirm that they can visualize relationship between the coefficients of a quadratic function and its minimum or maximum value, 20% of students agreed that they can verify the solutions of quadratic equations. With 50.7% of the students saying they enjoy using the GeoGebra app for studying mathematics and that they can use the GeoGebra software to learn mathematics at their own pace, 53.3% of the students said the GeoGebra software has increased their interest in learning mathematics. 42.7% and 46.7% of students agreed with the preceding statement, and 42.7% of students said that using the GeoGebra software has improved their understanding of how quadratic functions are used in real-world situations.

However, the data also highlight areas of concern. A considerable number of students express views that they are not able to use the GeoGebra

software effectively in mathematics. For instance, 45.3% of students cannot use the GeoGebra software to verify the solutions of quadratic equations, and 58.6% of respondents cannot use the GeoGebra software to investigate changes in the coefficients of quadratic functions, reflecting students not being exposed to the GeoGebra application. In contrast to this view of the students, 56.0% of the student wanted that GeoGebra software be integrated into the mathematics curriculum and 54.7% of them would recommend the use of the GeoGebra software to other students. 54.7% of the students also believe that GeoGebra software should be used in other subject areas beyond mathematics.

Discussion Related to Research Question.

The findings of this study have added to the fact that GeoGebra is a valuable educational tool that enhances students' understanding of mathematical concepts like maximum and minimum values of quadratic functions. Helping students to visualize and manipulate quadratic graphs in real-time, bridging the gap between theory and practical understanding.

The students' positive feedback on integrating GeoGebra into teaching demonstrates their acceptance of the software and its impact on their understanding of quadratic functions. Using GeoGebra motivates students to interact with quadratic functions and increases their confidence in solving related problems. Hanus and Fox (2015) suggest that increased interaction with teachers during lessons leads to more enjoyable learning experiences, which is supported by the majority of students' responses in this study indicating that they enjoyed lessons when GeoGebra was used.

Students' response that GeoGebra instruction helped them visualize the relationship between the coefficients of a quadratic function and its minimum

or maximum aligns with the findings of Shadaan and Leong (2013) and Pittalis and Christou (2010). These studies suggest that learners' ability to visualize concepts related to quadratic functions can enhance their performance in achievement tests.

In addition, the study's findings regarding how GeoGebra affected students' interest in mathematics are consistent with a 2019 study by Ling and Luan Wong regarding the relationship between interest and mathematical achievement in technology-enhanced learning.

In summary, from Table 1.3, students' attitude, motivation, and acceptance of the GeoGebra instruction shows positive responses, as they agree with the statement that GeoGebra is an essential tool for learning Mathematics and ought to be integrated into the mathematics curriculum. However, there are some negatives, such as not being able to use the GeoGebra application on their own to perform various operations while learning, showing students' lack of knowledge and exposure to the GeoGebra software.

Chapter Summary

The study's findings show that GeoGebra can help pupils improve quadratic function significantly. The findings showed that pupils performed better on the post-test than they did on the pre-test. Examining the effects of GeoGebra software on senior high school pupils was the goal of the study. accomplishment in figuring out quadratic functions' lowest and greatest values. These findings suggest that GeoGebra can serve as a valuable tool for enhancing students' understanding and proficiency in determining the minimum and maximum values of quadratic function. The study' result contribute to the broader discourse regarding the role of technological tools in teaching and

learning to help overcome challenges in mathematics education, reinforcing the importance of technological tools like GeoGebra, which can help students achieve educational success.

CHAPTER FIVE SUMMARY, CONCLUSIONS, AND RECOMMENDATION

This chapter provides a summary of the research, focusing on the study's purpose, methodology, and findings. It also presents conclusions based on the study's findings and offers recommendations to emphasize the significance of integrating GeoGebra software into mathematics education.

Summary

The purpose of this study was to look into how high school students' use of the GeoGebra program affected their capacity to find the minimum and maximum values of quadratic functions. The following were the research topic and hypothesis:

Research Question: What is the effect of using the GeoGebra software on the achievement of senior high school students in determining the minimum and maximum values of a quadratic function?

Hypothesis H_0 : There is no statistically significant difference in the mathematics achievement test scores of students taught using the GeoGebra software and the same students taught using the conventional method.

This study investigated the causes and effects of incorporating GeoGebra software into the teaching of the minimum and maximum values of quadratic functions. It used a quasi-experimental design and focused on the same group of students. The two schools and classes participating in the Cape Coast city, in Ghana's central area, were identified by purposeful sampling. Seventy-five students from these schools participated in the study. Pre-test, post-test, research questionnaire, and GeoGebra teaching as an intervention comprised the process. The study question and hypothesis were addressed by

the analysis of pre- and post-test data. Additionally, the research question was reinforced by the questionnaire responses, which demonstrated the participants' favorable attitudes, motivation, and acceptance of GeoGebra education. Frequencies, percentages, means, standard deviations, and t-tests were utilized in the analysis; percentages were also used to assess the questionnaire data

Findings

The study identified three main findings, which are outlined below:

1. Students taught with the GeoGebra software performed better on average compared to those taught using the Traditional method.
2. Students taught with the GeoGebra program and the same students taught using the traditional method showed statistically significant differences in their mathematics achievement test scores.
3. Students' confidence, motivation, and attitude toward learning and resolving quadratic function problems were all improved by the GeoGebra program.

Conclusion

Three conclusions were derived from the study's findings:

Firstly, the study indicated that GeoGebra, as a teaching tool, enhances students' academic performance in mathematics. This implies that high school students perform better in mathematics when taught with GeoGebra compared to traditional teaching methods.

Also, the findings revealed that, by incorporating GeoGebra into mathematics instruction, students can enhance their comprehension of concepts and problem-solving skills. As a result, students show notable progress in their

academic performance due to their ability to identify the connections between the minimum and maximum values of quadratic functions in mathematics.

Additionally, incorporating GeoGebra into teaching and learning helps to inspire students to engage with mathematics more effectively. Utilizing GeoGebra as part of a blended instructional approach can enhance students' enthusiasm for the subject.

The study's results indicated that utilizing innovative methods like integrating GeoGebra into education has been more successful than traditional teaching approaches. This contributes to the existing body of research supporting the effectiveness of GeoGebra software in enhancing students' comprehension of mathematical concepts

Recommendations

The Ghana Education Service and other stakeholders are suggested to consider the following recommendations in light of the study's findings:

1. Introducing pre- and in-service teachers to GeoGebra as a cutting-edge approach to teaching mathematics can greatly enhance senior high school students' academic achievement. This introduction can be made easier by workshops and seminars led by the Ghana Education Service and other education stakeholders, which will improve the tactics and abilities of math teachers.
2. The integration of computers and engaging instructional software, such as GeoGebra, into the Senior High School Mathematics curriculum can help teachers facilitate students' understanding of quadratic ideas and improve their ability to solve problems. It is anticipated that the pupils' performance will significantly increase as a result of this approach.

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APPENDICES

APPENDIX A PRE-TEST

1. a) Copy and complete the table of values for the relation $Y = 2x^2 - x - 2$,

for $-4 \leq x \leq 4$.

X	-4	-3	-2	-1	0	1	2	3	4
Y		19			-2				26

b) Using a scale of 2cm to 1unit on the X-axis and 2cm to 5 units on the Yaxis, draw the graph of $Y = 2x^2 - x - 2$, for $-4 \leq x \leq 4$.

c) From the graph, find the minimum value of Y.

APPENDIX B POST- TEST

1. a.) Copy and complete the table of values for the relation $Y = -x^2 +$

$x + 2$, for $-3 \leq x \leq 3$.

X	-3	-2	-1	0	1	2	3
Y		-4		2			-4

b) Using a scale of 2cm to 1unit on the X-axis and 2cm to 2 units on the Y-axis, draw the graph of $Y = -x^2 + x + 2$, for $-3 \leq x \leq 3$.

c) Find the maximum value of y from the graph.

APPENDIX C QUESTIONNAIRE FOR STUDENTS

Dear Respondent,

This research questionnaire is designed to investigate the "Effect of GeoGebra App on SHS Student Achievement in Determining Minimum and Maximum Values of a Quadratic Equation in Cape Coast Metropolis." Your responses will be kept confidential and used solely for academic purposes. Please answer the questions to the best of your knowledge.

SECTION A: DEMOGRAPHICS OF RESPONDENTS

About the following details, tick (✓) the response that best describes you or provide short response when required.

1. Gender

i. ☐ Male ii. ☐

☐ Female

2. Age

☐ Below 15years

☐ 15years - 18years

☐ 19years - 22years ☐

23years and above

3. Programme/Course of Study.

.....

4. Access to a smartphone or computer at home:

Yes ☐

No ☐

5. Mathematics Performance Level: using your last examination score in mathematics, tick (✓) the option using the scale below

Below average (0 to 49), Average (50 to 70), and above average (71 to 100).

Below Average ☐

Average ☐

Above Average ☐

Instructions for Section B:

This section explores your experience with using the GeoGebra app. Read the instructions below regarding GeoGebra and tick (✓) your experiences

	Statement	Strongly Agree	Agree	Disagree	Strongly Disagree
6.	I have used the GeoGebra app before.				
7.	I find the GeoGebra app easy to use				
8.	I use the GeoGebra app regularly for my mathematics learning				
9.	The GeoGebra app has helped me better understand mathematical concepts.				
10.	I feel confident in my ability to use the GeoGebra app for my mathematics learning				
11.	I have received training or guidance on how to use the GeoGebra app effectively.				
12.	The GeoGebra app provides me with timely feedback and support when I use it.				
13.	I find the GeoGebra app's features and tools to be comprehensive and helpful.				

Section C: Quadratic Functions and GeoGebra

This section focuses on your understanding of quadratic functions and how GeoGebra has influenced your learning. Read the instructions below regarding GeoGebra and tick (✓) your experiences.

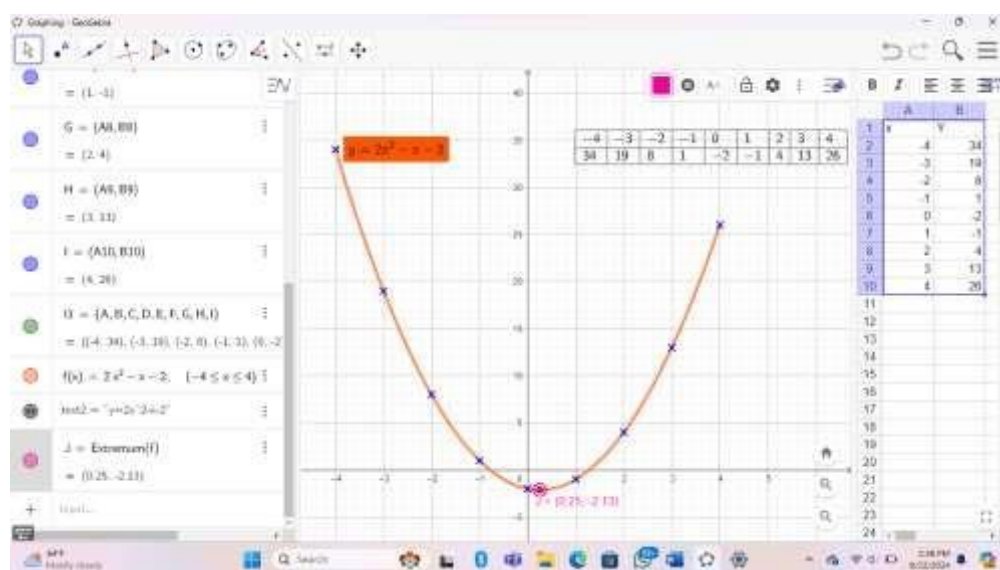
	Statement	Strongly Agree	Agree	Disagree	Strongly Agree
14.	Before using GeoGebra, I was confident in determining the maximum and minimum values of a quadratic function?				
15.	After using GeoGebra, I am confident in determining the maximum and minimum values of a quadratic function.				
16.	GeoGebra has helped improve my understanding of the concept of maximum and minimum values of a quadratic function.				
17.	GeoGebra has improved my ability to solve problems related to maximum and minimum values of a quadratic function.				
18.	The GeoGebra app has helped me visualize the relationship between the coefficients of a quadratic function and its minimum or maximum value.				
19.	I can use the GeoGebra app to verify the solutions of quadratic equations.				
20.	I can use the GeoGebra app to investigate how changes in the coefficients of a quadratic function affect its minimum or maximum value.				
21.	The GeoGebra app has helped me better understand the realworld applications of quadratic functions				

Section D: Overall Assessment

This section seeks your general opinion about using GeoGebra for mathematics learning. Read the instructions below regarding GeoGebra and tick (✓) your experiences.

	Statement	Strongly Agree	Agree	Disagree	Strongly Disagree
22.	I enjoy using the GeoGebra app for my mathematics learning.				
23	The GeoGebra app has made me more interested in learning mathematics.				
24.	I find the GeoGebra app to be a useful tool for learning mathematics				
25.	The GeoGebra app has made me more interested in learning mathematics				
26.	The GeoGebra app has helped me learn mathematics at my own pace				
27.	I would recommend the use of the GeoGebra app to other students.				
28.	The GeoGebra app should be integrated into the mathematics curriculum				
29.	I believe the GeoGebra app should be used in other subject areas beyond mathematics				

APPENDIX D MARKING SCHEME FOR PRE-TEST



TableB3

Graph.....B3

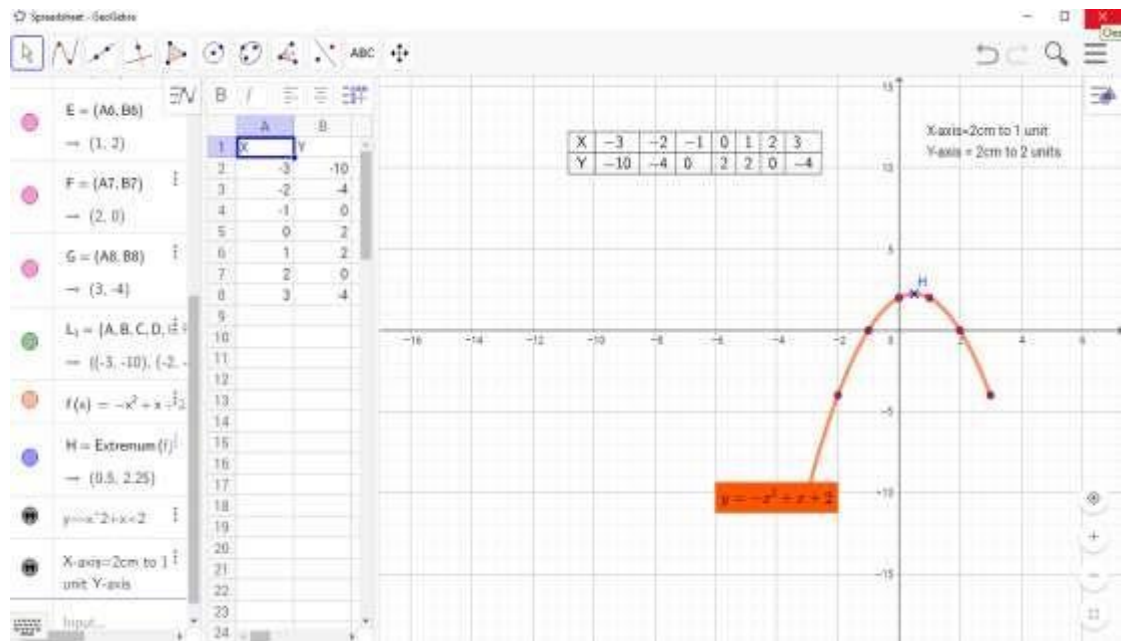
Scale..... A1

Axes.....A1

Maximum Value.....A2

Total marks = 10

APPENDIX E MARKING SCHEME FOR POST-TEST



TableB3

Graph.....B3

Scale..... A1

Axes.....A1

Maximum Value.....A2

Total marks = 10

APPENDIX F INTRODUCTORY LETTER TO SCHOOL

UNIVERSITY OF CAPE COAST
COLLEGE OF EDUCATION STUDIES
FACULTY OF SCIENCE AND TECHNOLOGY EDUCATION

TELEPHONE:
E-MAIL: dmcte@ucc.edu.gh
WEBSITE: www.ucc.edu.gh



Department of Mathematics and ICT
Education
University Post Office
Cape Coast

Your Ref:
Our Ref: DMCTE/U.I/V.1/276
Date: 22nd August, 2024

TO WHOM IT MAY CONCERN:

Dear Sir / Madam,

INTRODUCTORY LETTER

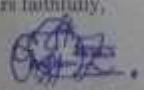
We write to confirm that the under-listed students with their respective registration numbers are level 400 B.Ed. (Mathematics Education) of the Department of Mathematics and ICT Education, Faculty of Science and Technology Education, University of Cape Coast.

Name	Registration Number
1. Margaret Baidoo	ET/MAT/20/0193
2. Emmanuel Angmorley	ET/MAT/20/0086
3. Philip Kumah Mintah	ET/MAT/20/0209
4. Akpfo Ebenezer Selorm	ET/MAT/20/0070

They are currently undertaking their project work on the topic "EFFECT OF GEOGEBRA APP ON STUDENTS ACHIEVEMENT IN DETERMINING MINIMUM AND MAXIMUM VALUES OF A QUADRATIC EQUATION AT THE SENIOR HIGH SCHOOL LEVEL." and would need information that will enable them to continue with their project.

We would, therefore, be grateful if they were given the needed assistance to support their studies.

Yours faithfully,


Dr Christopher Yarkwah
SUPERVISOR


ST. AUGUSTINE'S COLLEGE
CAPE COAST